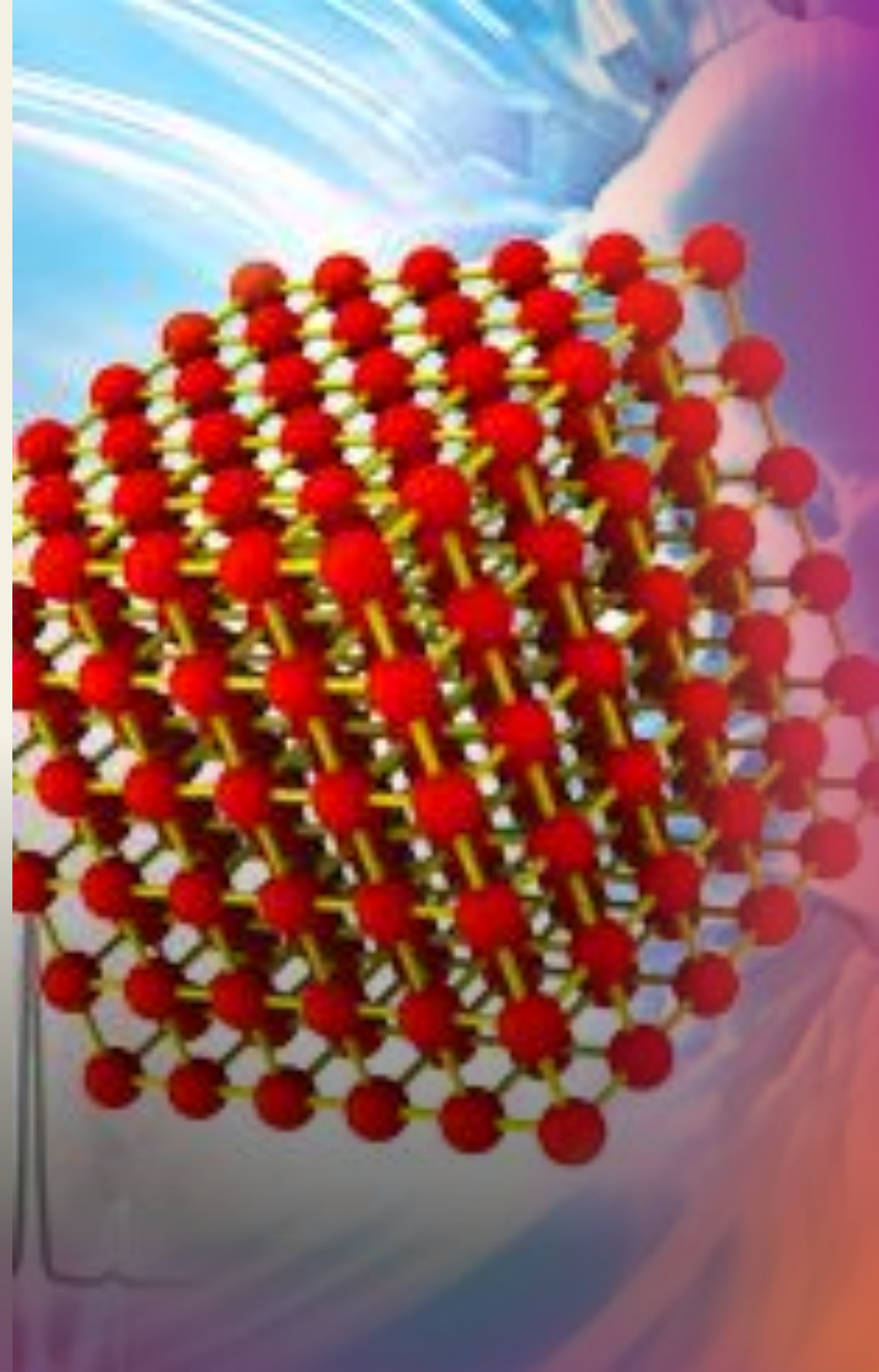


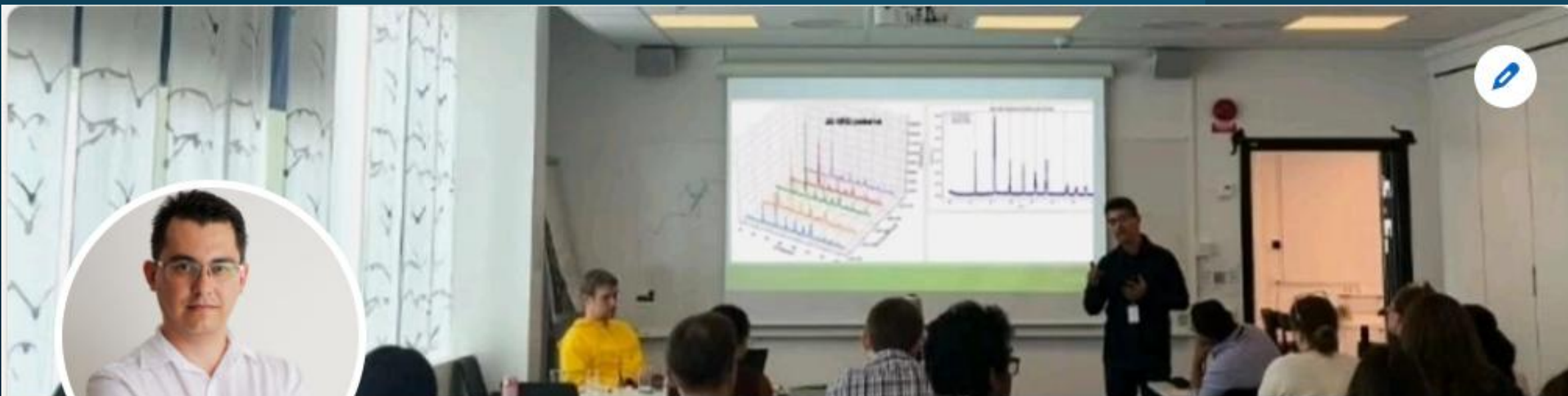
XXXII SEMANA DA FÍSICA

DO QUÂNTICO AO
COMPUTACIONAL:

Fronteiras
da Física,
Tecnologia
e Educação

INSCRIÇÕES: 10 a 21 de
novembro de 2025.
SEM TAXA DE INSCRIÇÃO





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Department of Physics, Lund University



Faroe Islands

Norway

Sweden

Finland

Oslo

Stockholm

Helsinki

Tallinn

St Petersburg
Санкт-Петербург

Edinburgh

United Kingdom

Liverpool

London

Amsterdam

Netherlands

Brussels

Belgium

Cologne

Luxembourg

Paris

Germany

Hamburg

Denmark

Lund

Poland

Bel

Prague

Baltic Sea

Gulf of Bothnia

Estonia

Riga

Latvia

Lithuania

Vilnius

Minsk

Мінск

Belarus

Moscow

Москва

Nizhny Novgorod
Нижний Новгород

Voronezh

Воронеж

Kyiv

Київ

Kharkiv

Харків

Volgogr

Волгоград

Saratov

Саратов





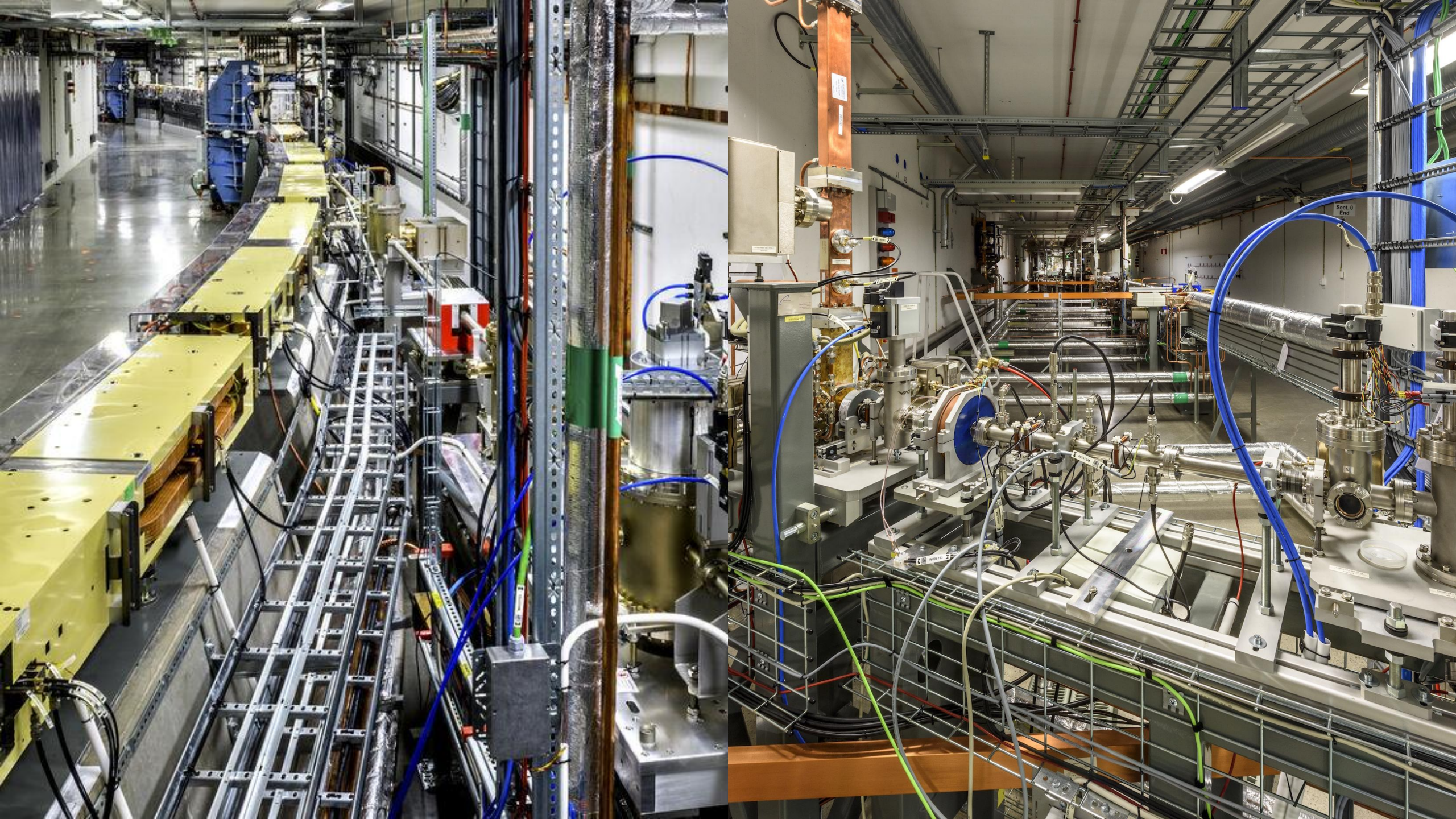




*Anne L'Huiller prêmio Nobel
de Física 2023*











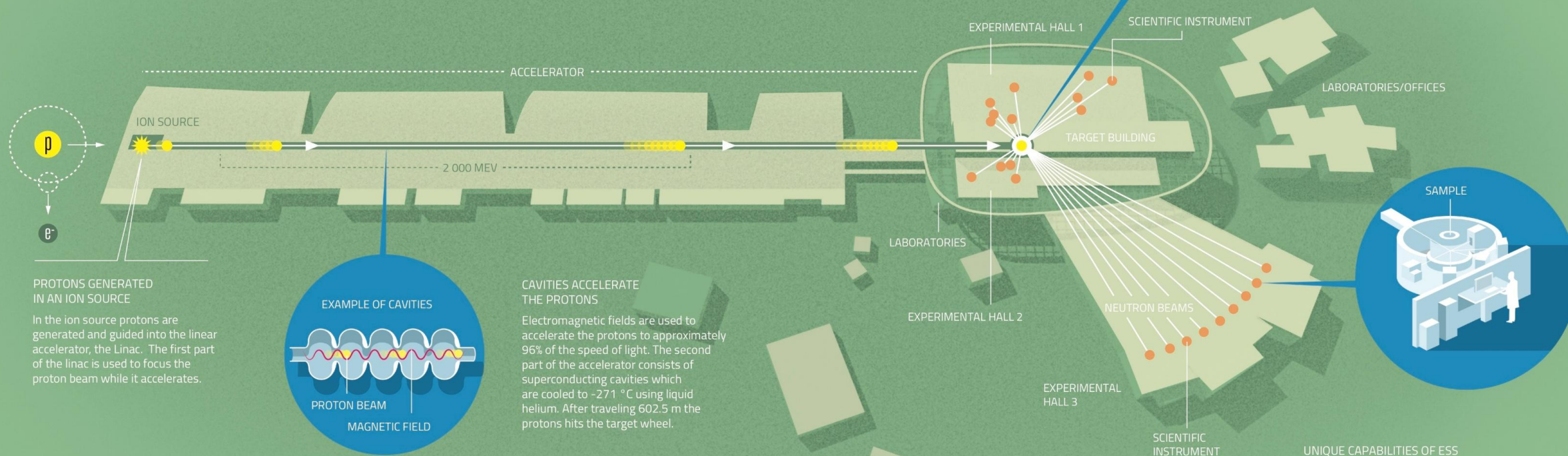
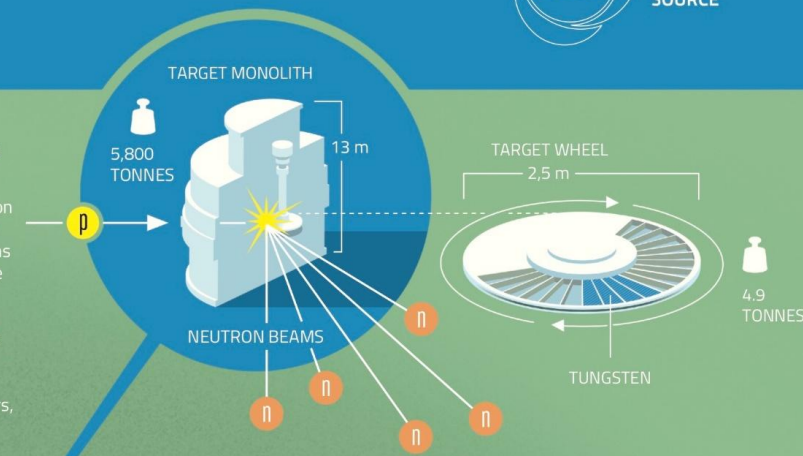
European Spallation Source

The European Spallation Source (ESS) is a multi-disciplinary research centre based on the world's most powerful neutron source. ESS will give scientists new possibilities in a broad range of research, from life science to engineering materials, from heritage conservation to magnetism. ESS is a pan-European project, with Sweden and Denmark serving as host countries. The main research facility is being built in Lund, Sweden, and the Data Management and Software Centre (DMSC) is located in Copenhagen, Denmark.



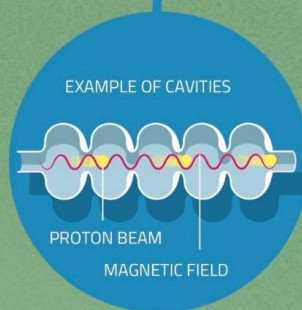
THE TARGET IS THE NEUTRON SOURCE

When the accelerated protons hit the rotating tungsten target wheel spallation occurs and neutrons are scattered from the tungsten nucleus. The more neutrons produced and collected in the target, the "brighter" the neutron source. The neutrons are directed through moderators and neutron guides to the scientific instruments where they are used for experiments. The Target monolith consists of the Target wheel, moderators, cooling systems and shielding and weighs approximately 5,800 tonnes.



PROTONS GENERATED IN AN ION SOURCE

In the ion source protons are generated and guided into the linear accelerator, the Linac. The first part of the linac is used to focus the proton beam while it accelerates.



CAVITIES ACCELERATE THE PROTONS

Electromagnetic fields are used to accelerate the protons to approximately 96% of the speed of light. The second part of the accelerator consists of superconducting cavities which are cooled to -271°C using liquid helium. After traveling 602.5 m the protons hit the target wheel.

TOTAL BUILDING AREA 65 000 m²

The ESS facility will be approximately 650 metres in total length. The target building will be 125 metres long, and about 30 metres high. The 537-meter-long accelerator tunnel is built underground and will be

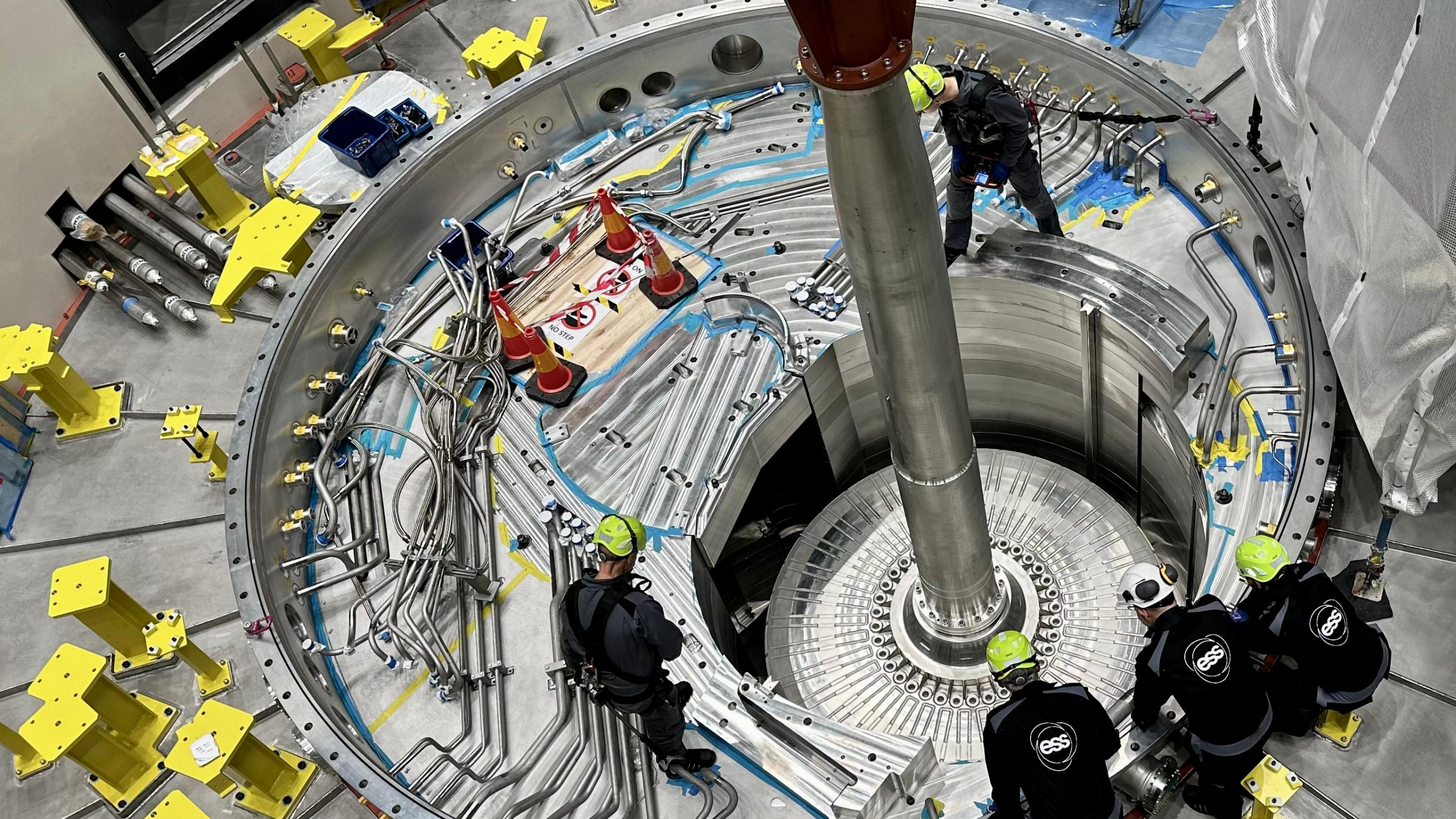
Concrete: 50 000 m³
Rebar: 6 000 tonnes
Pipes: 40 km
Cables: 2 000 km

PILES TO AVOID MOVEMENTS

The heavy Target building and experimental halls are resting on a total of 6,400 piles of different types, in order to avoid unwanted movements in the structure.

UNIQUE CAPABILITIES OF ESS

ESS will have 22 tailor-made instruments located in three experimental halls. Neutrons are excellent for probing materials on an atomic and molecular level – everything from motors and medicine, to plastics and proteins. The neutrons hit the sample and detectors register the neutron scattering, giving precise information about the material's structure and dynamics.







Create your space here

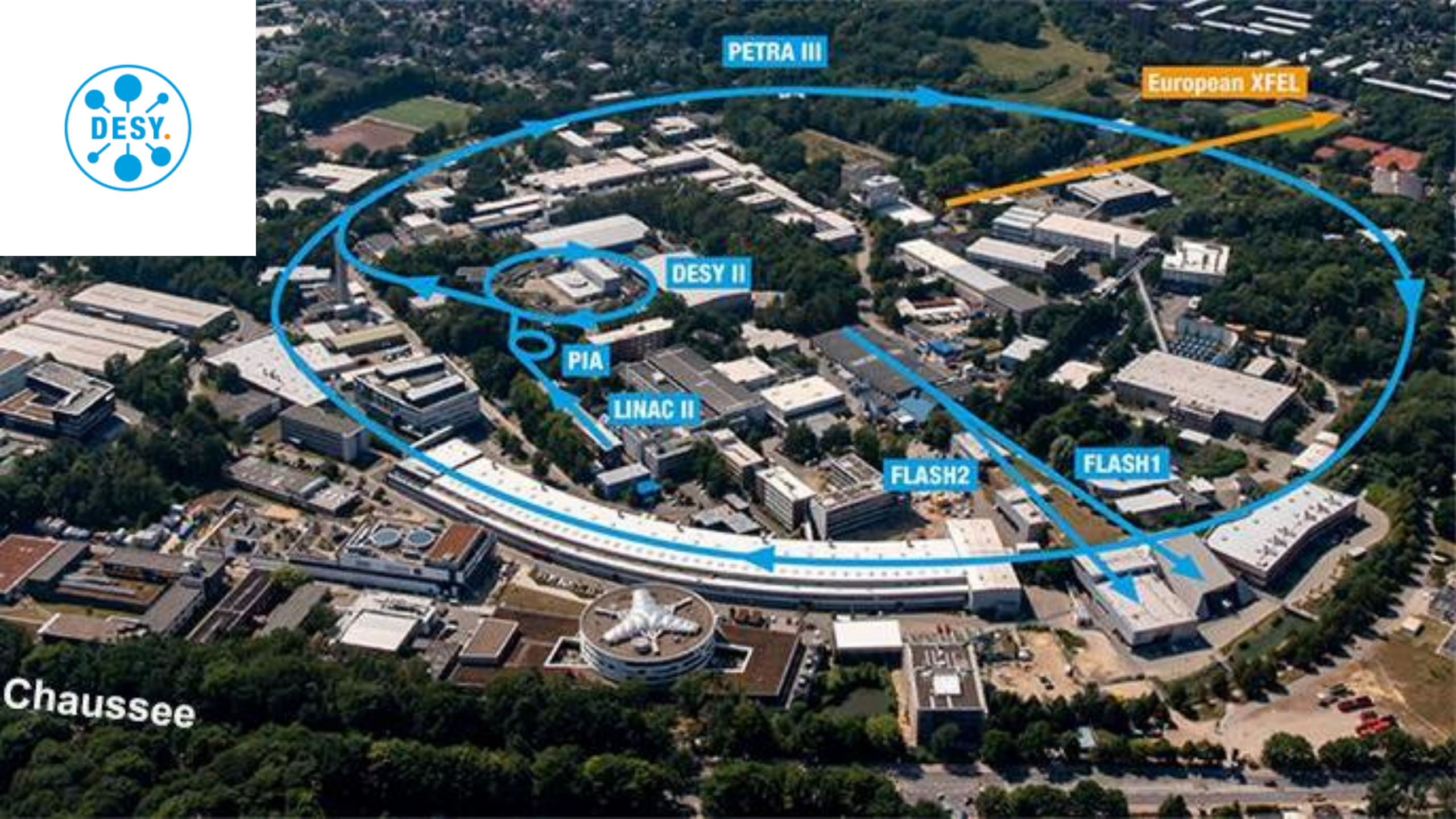


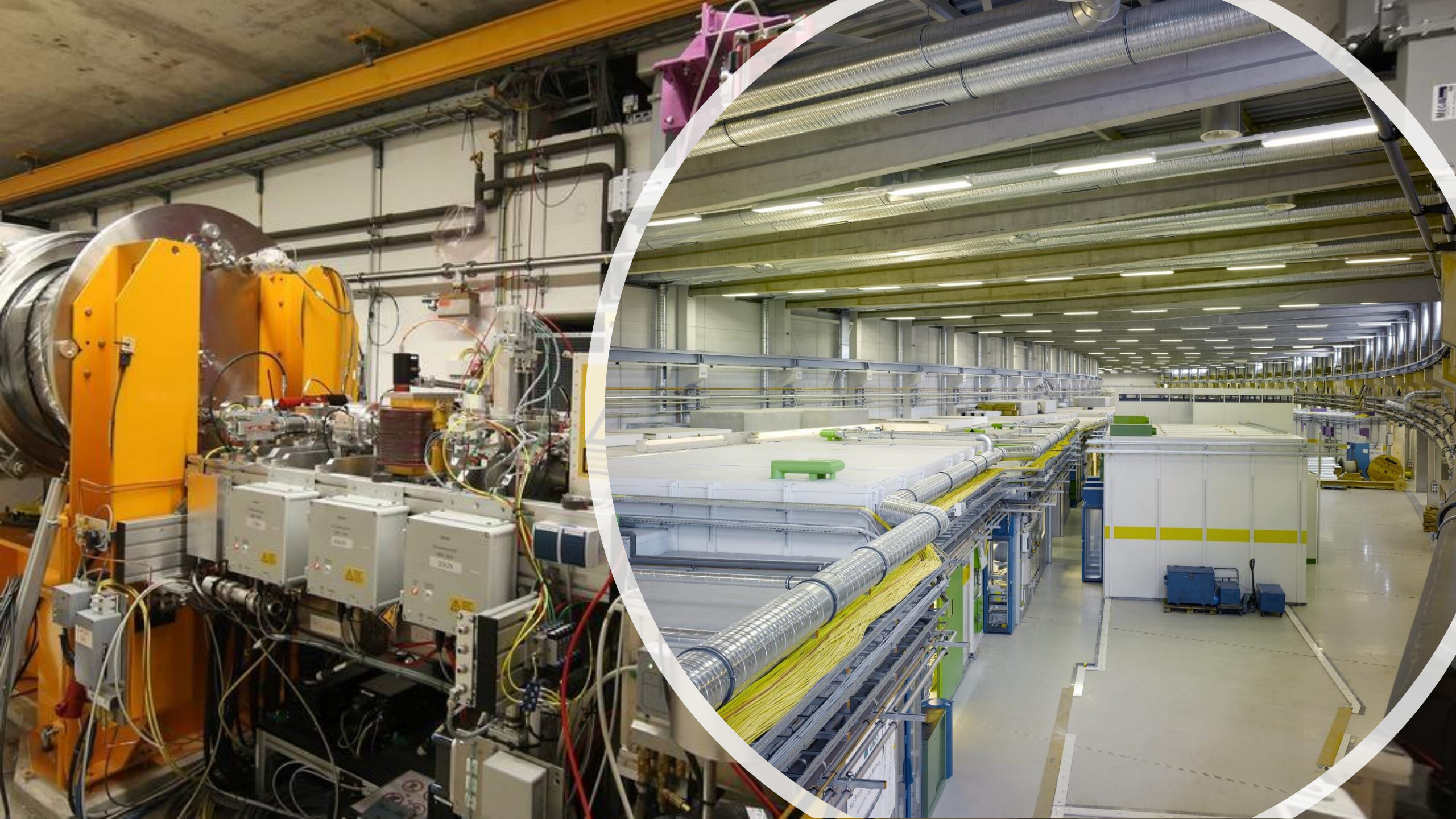
NNSP

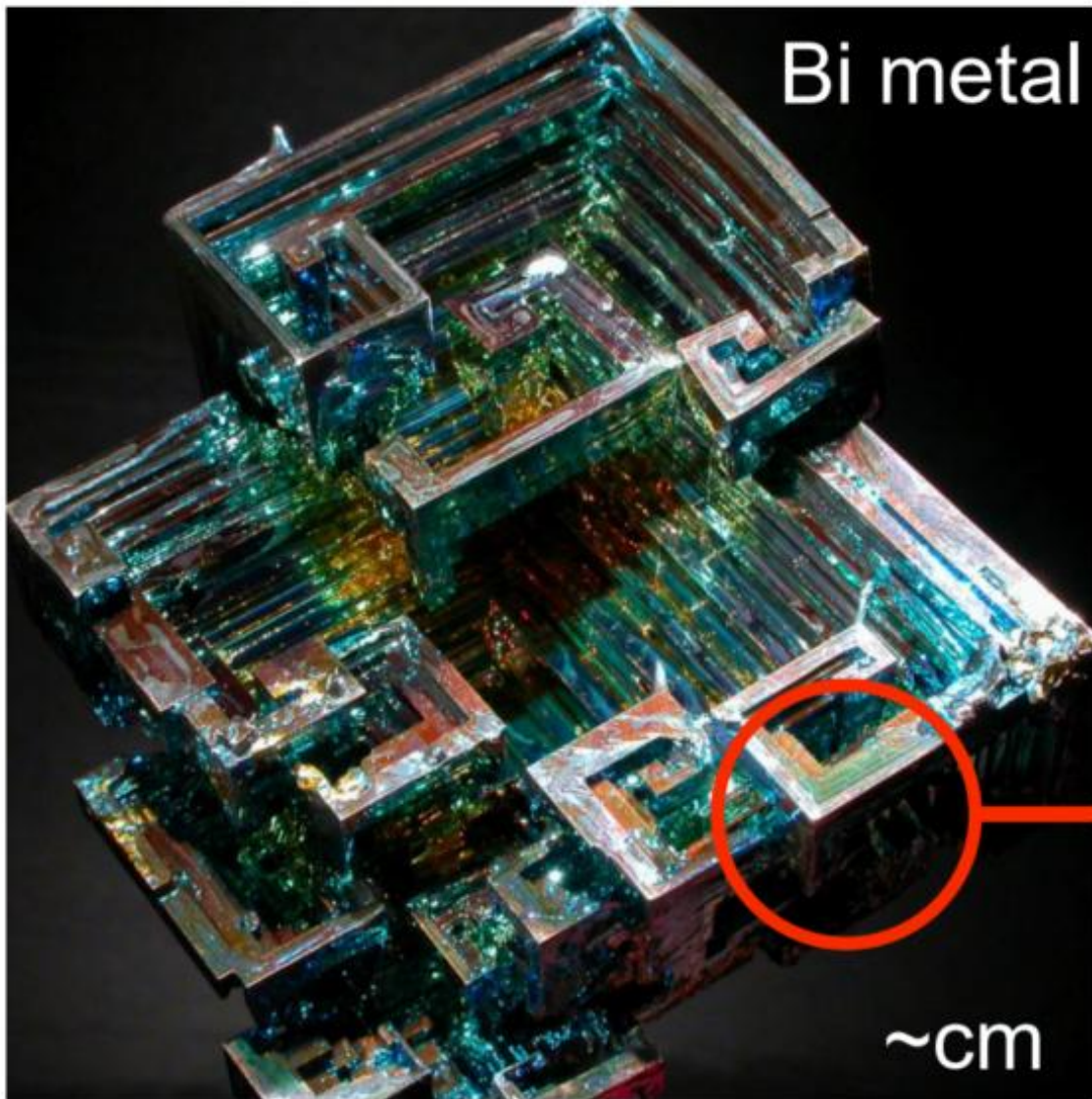


LINXS

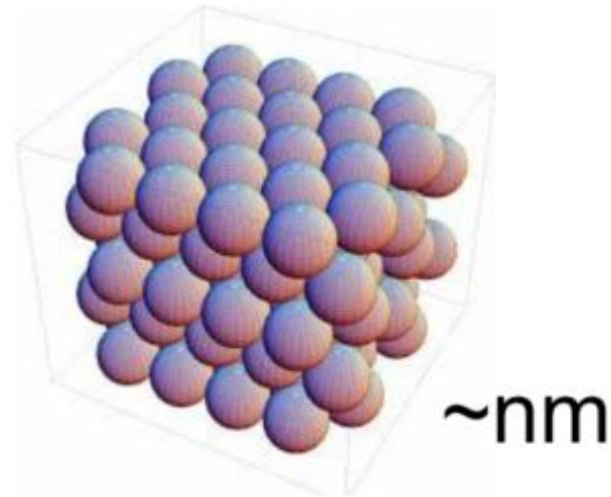
UWE





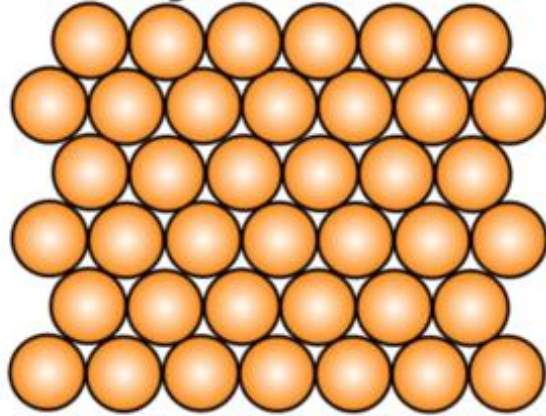


Crystals are homogeneous, anisotropic solid-state compound, which have **3D periodically ordered structures**.



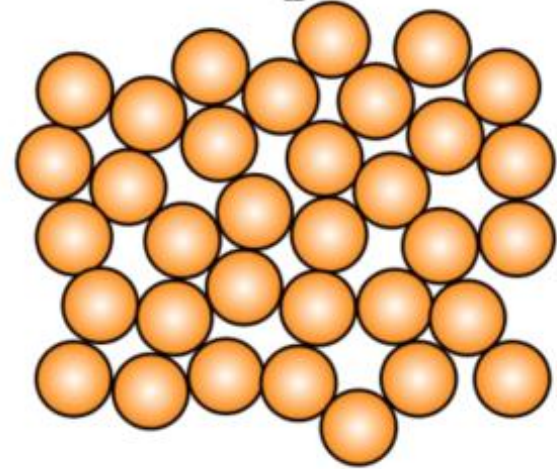
Solid state materials

Crystalline



Homogeneous, anisotropic solid-state systems, whose atoms are naturally arranged in periodically repeated symmetric patterns

Amorphous



Solid-state compound without a 3D periodic order. (gels, glasses, wood, plastic...)

Motivation

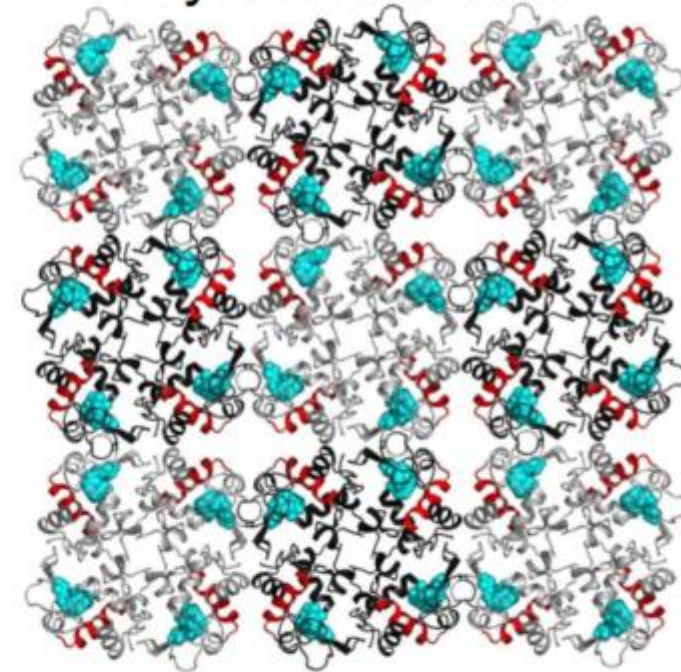
Motivation

Crystallography:

the study the periodic arrangement of atoms in solid condensed matter.

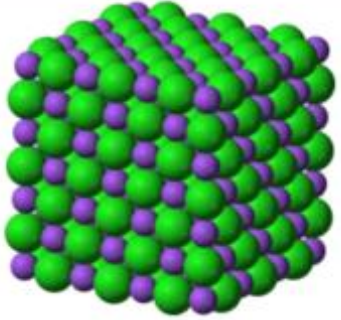


Crystal structure



How to describe a crystal structure in the simple terms of the discrete geometry that underlies many physical and spectroscopic properties of matter.

Crystal structure

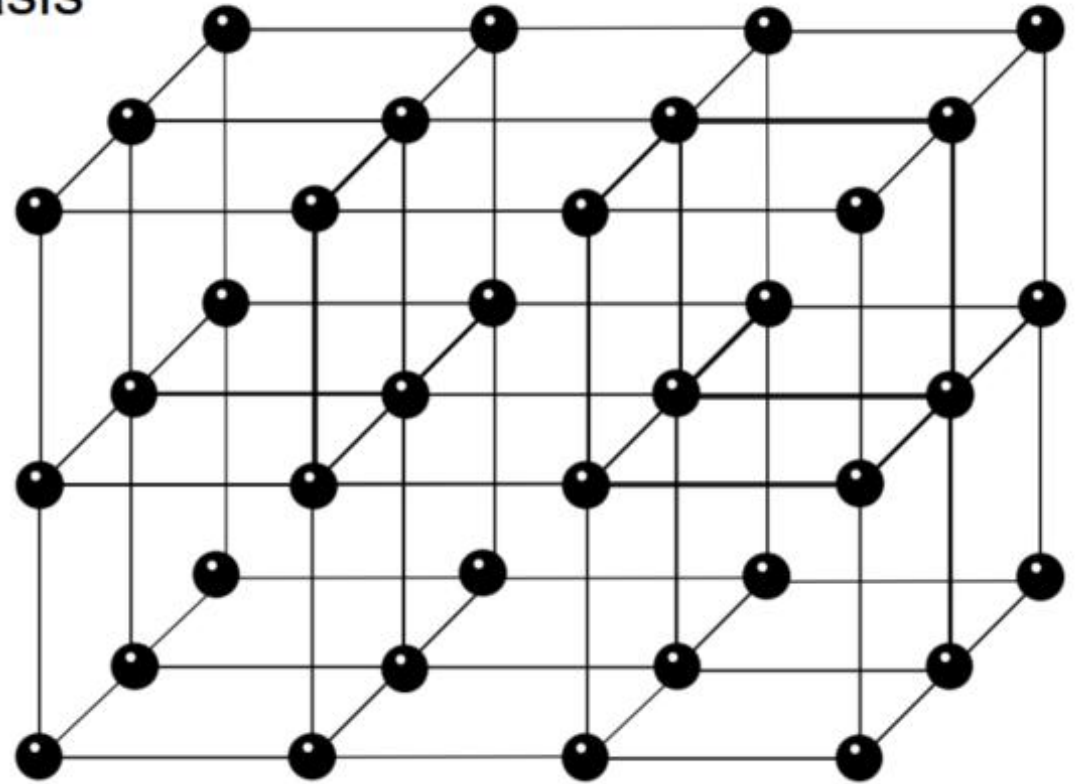


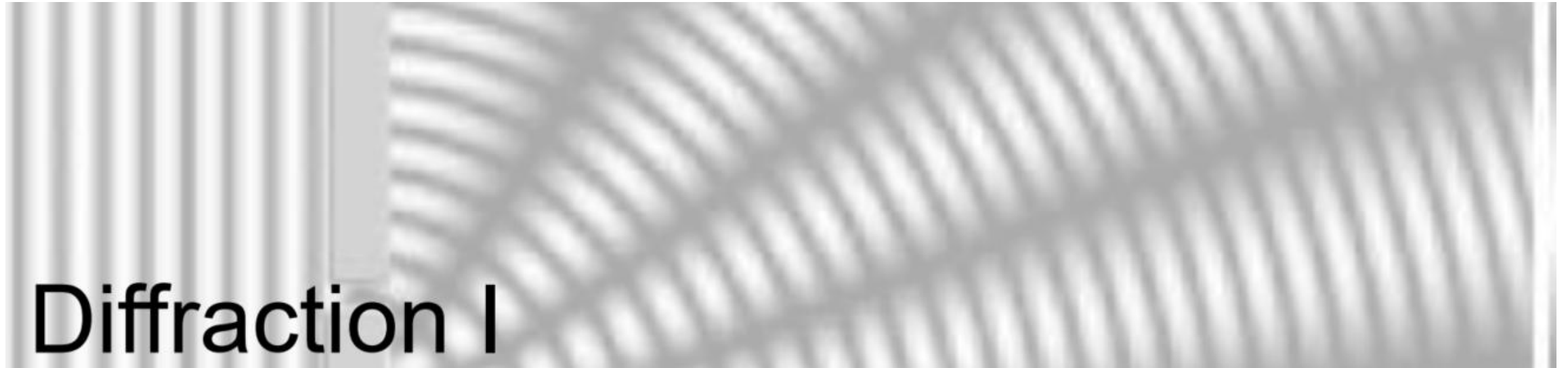
Crystal structure = Lattice + Basis



Infinite arrangement of points is space (3D)/ in the plane (2D) where all points have same surrounding.

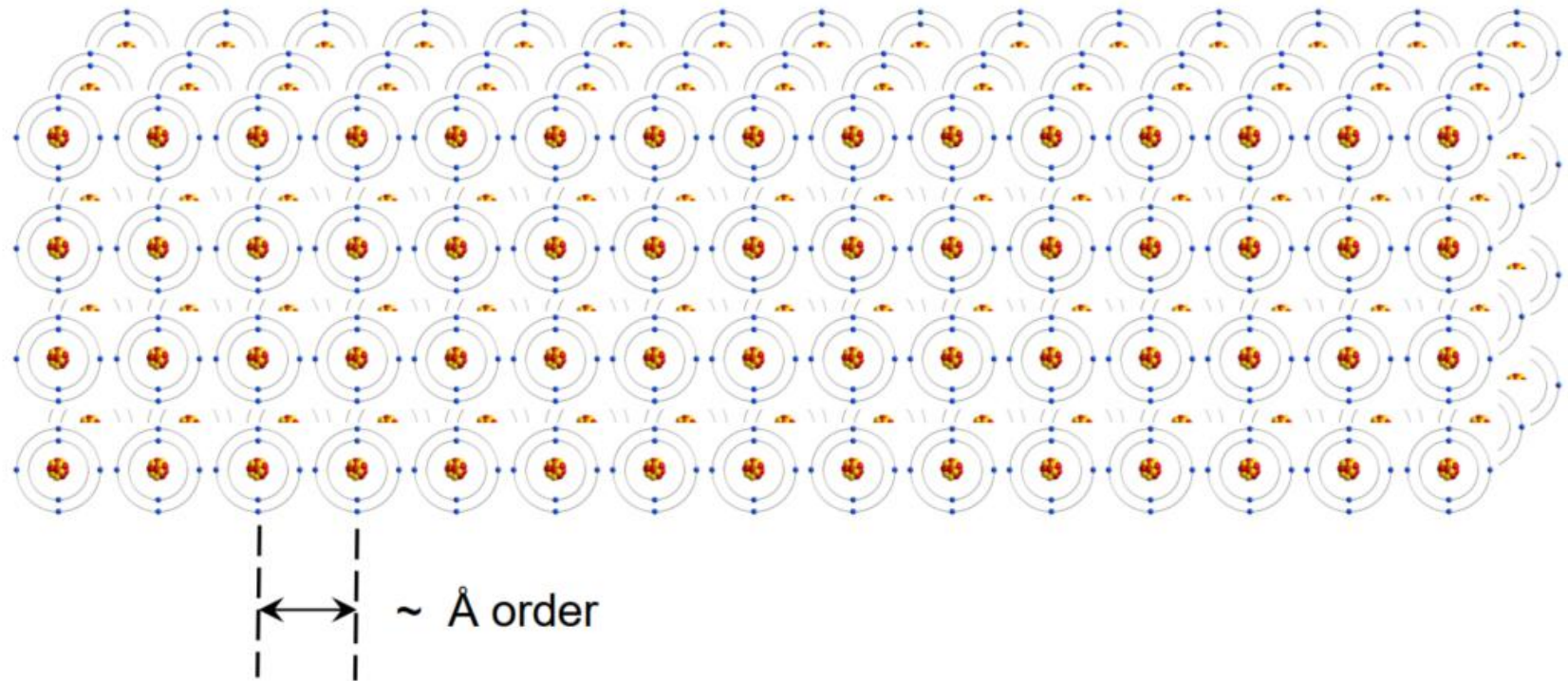
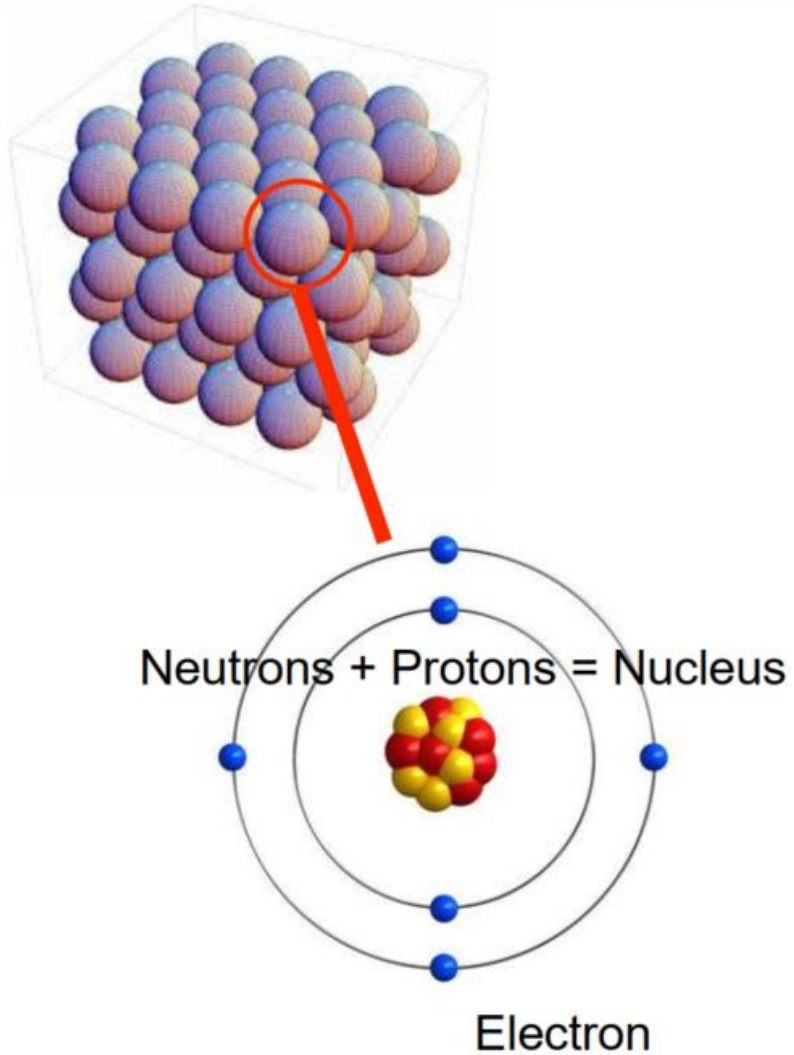
Surface/border points do not belong to the lattice!



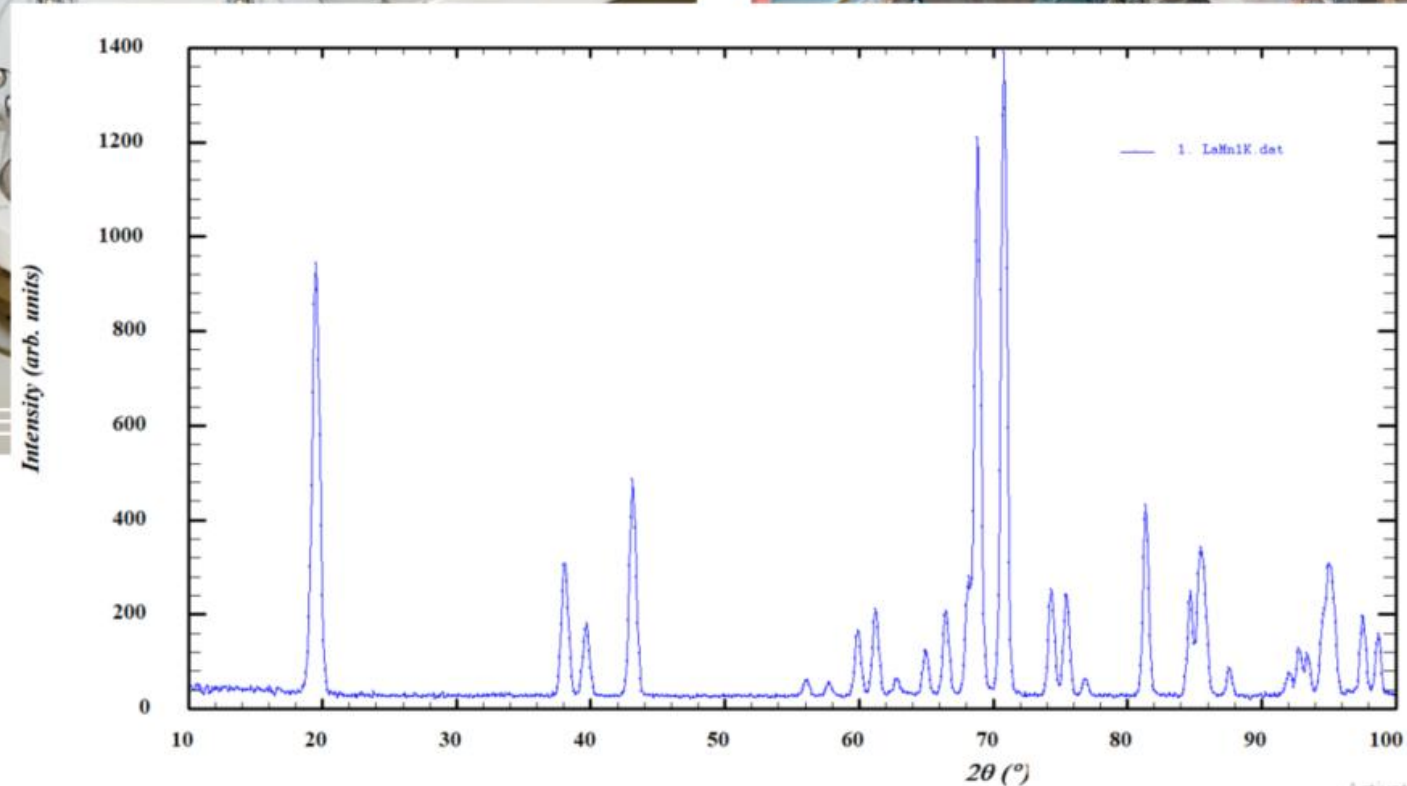
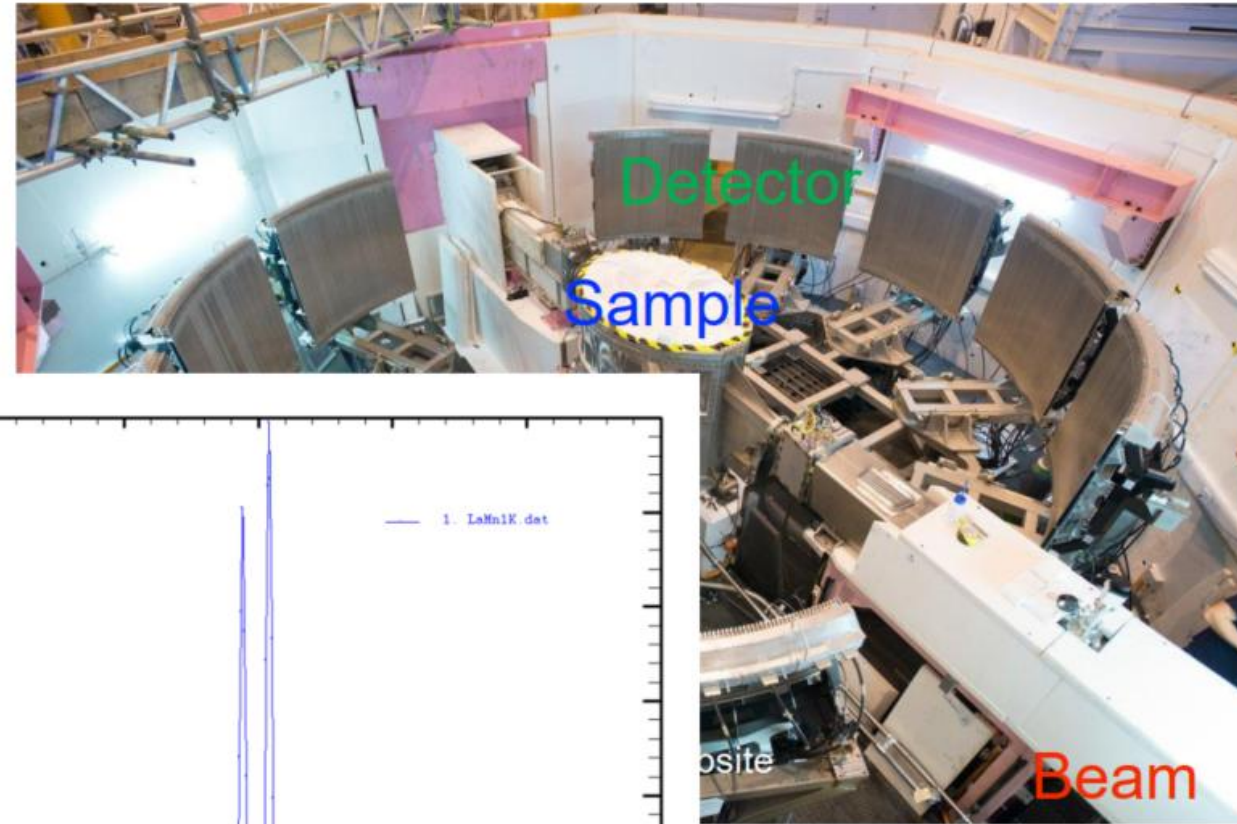
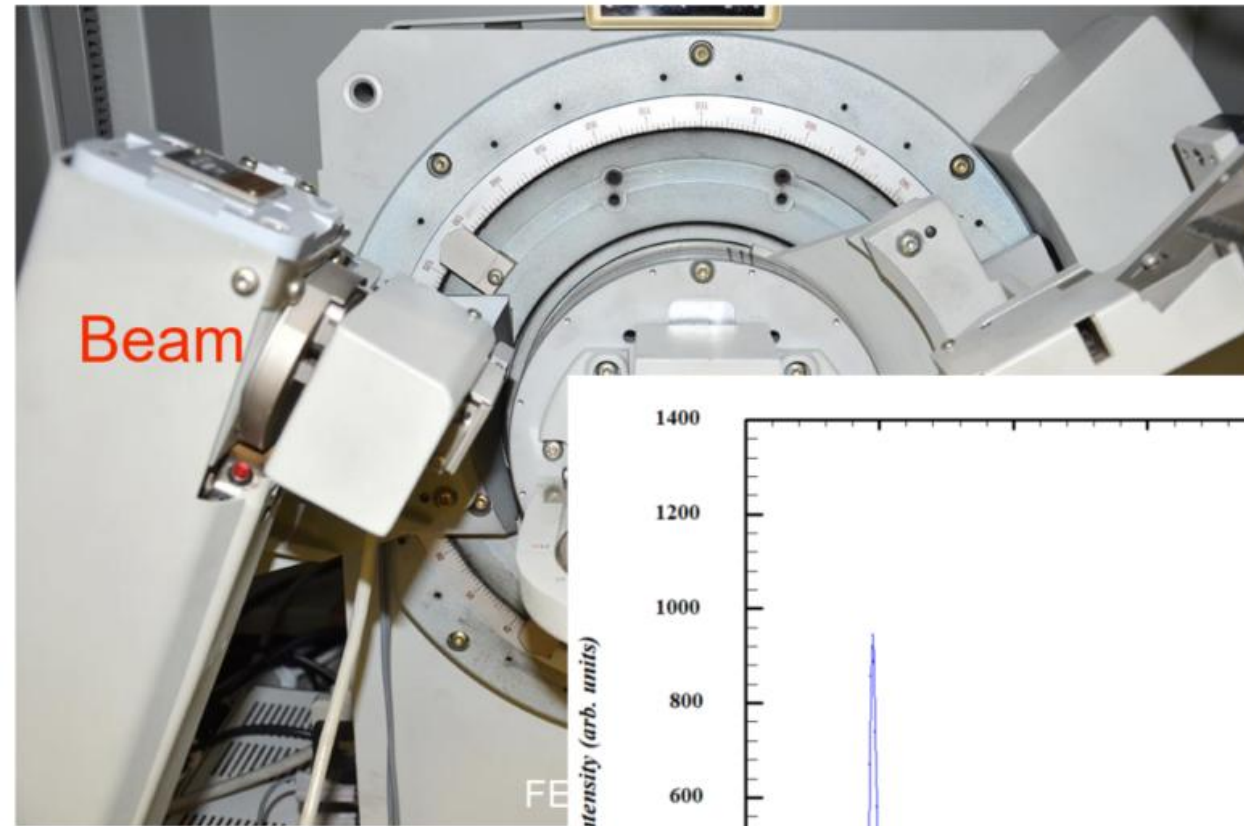


Diffraction I

Atoms



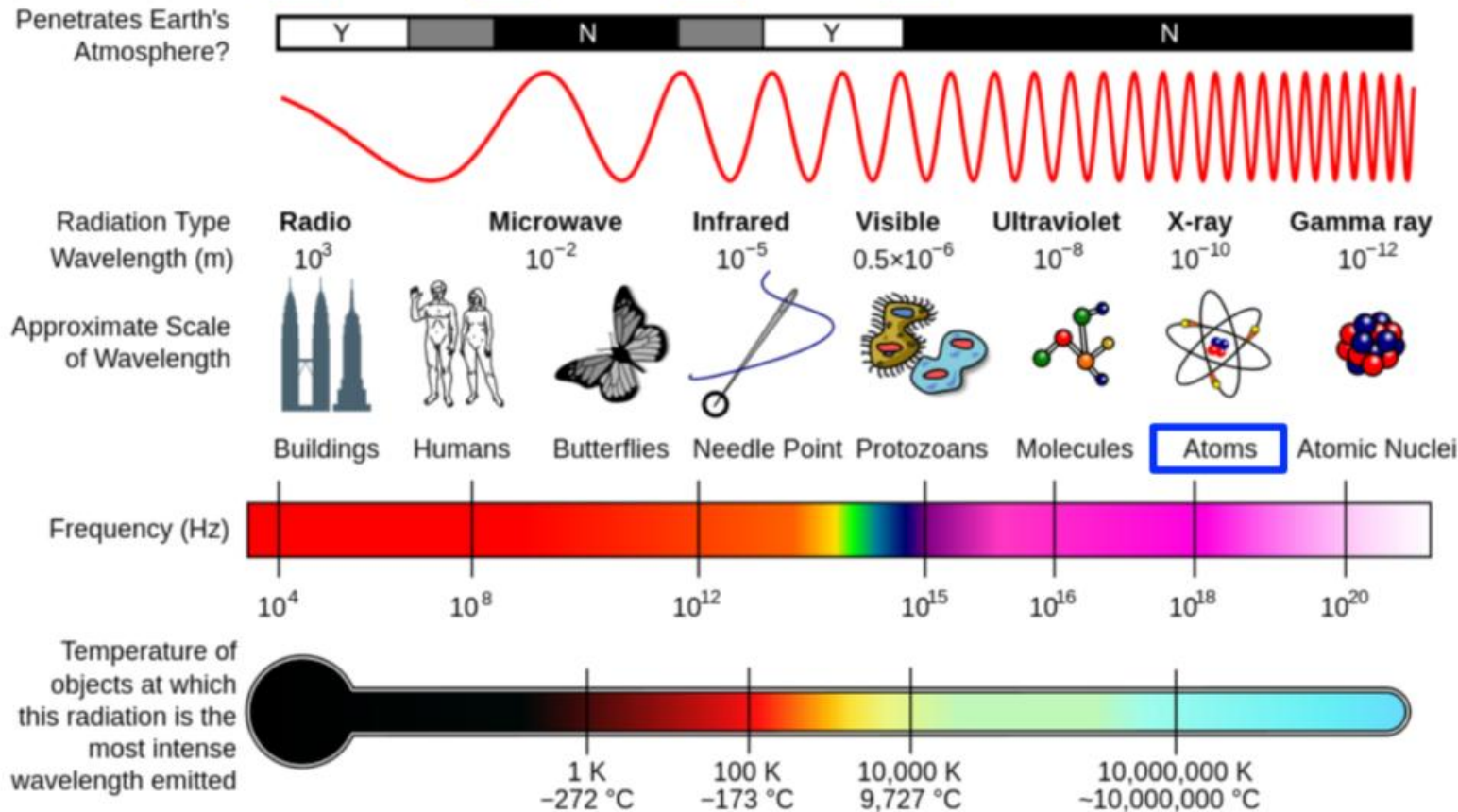
What is a diffraction experiment ?



Waves allow us to see the world



The wavelength λ is the spatial period of a periodic wave.



Crystal structure scale is comparable to the wavelength of X-ray, Neutrons.

$$\text{X-ray} \\ \lambda = 0.1 \sim 10 \text{ \AA}$$

$$\text{Neutron} \\ \lambda = 1 \sim 4 \text{ \AA}$$

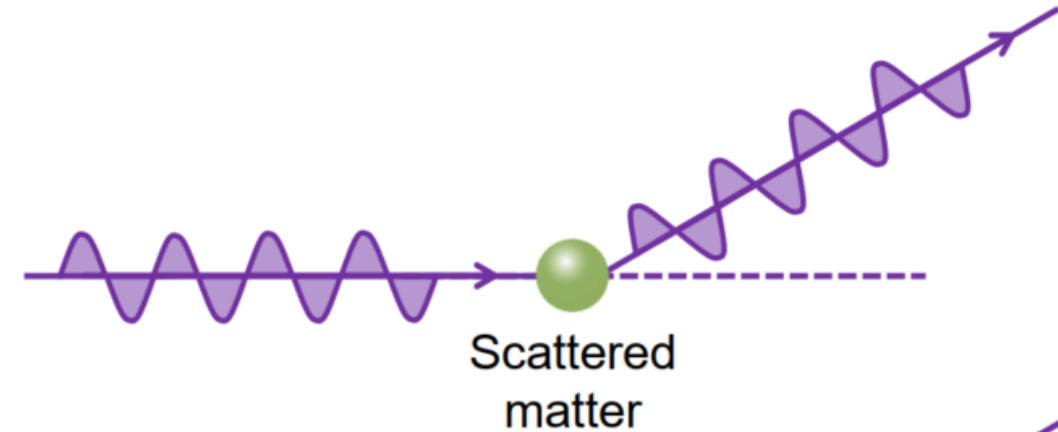
Atoms are visible by X-rays, electrons and Neutrons!!!

Neutronwaves show you...

(1) where the atoms and spins are in the material!

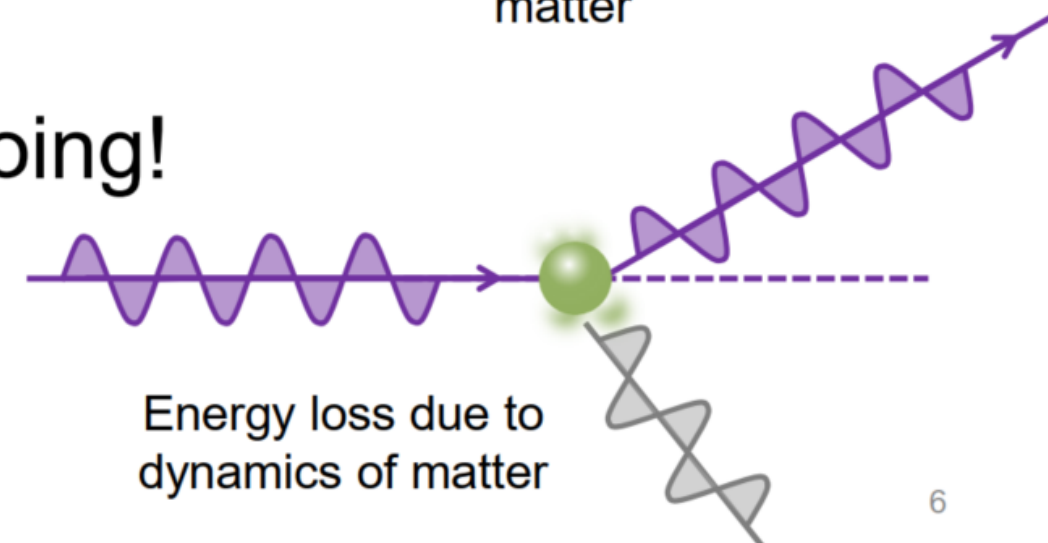
Elastic Scattering (Diffraction)

- 👉 Position of the atoms
- 👉 Spin configurations
- 👉 Microstructure of materials (e.g. Size and shape of grains, Preferential orientation)

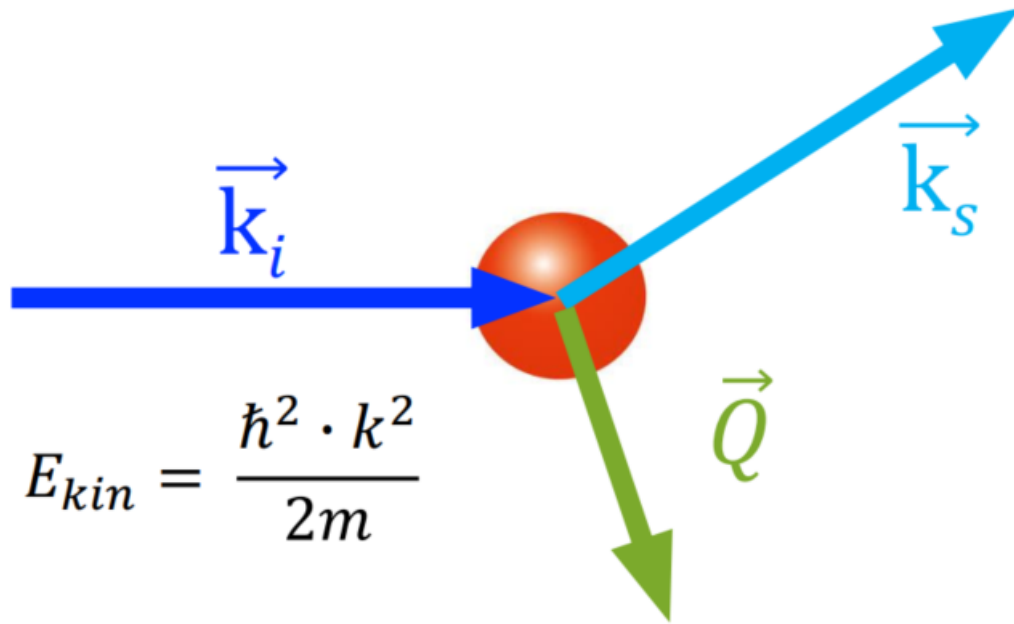


(2) what the atoms and spins are doing!

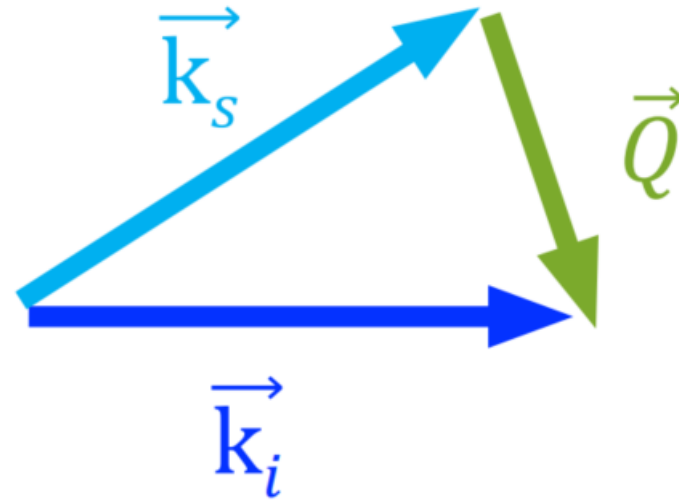
Inelastic Scattering



Scattering triangle



Scattering triangle:



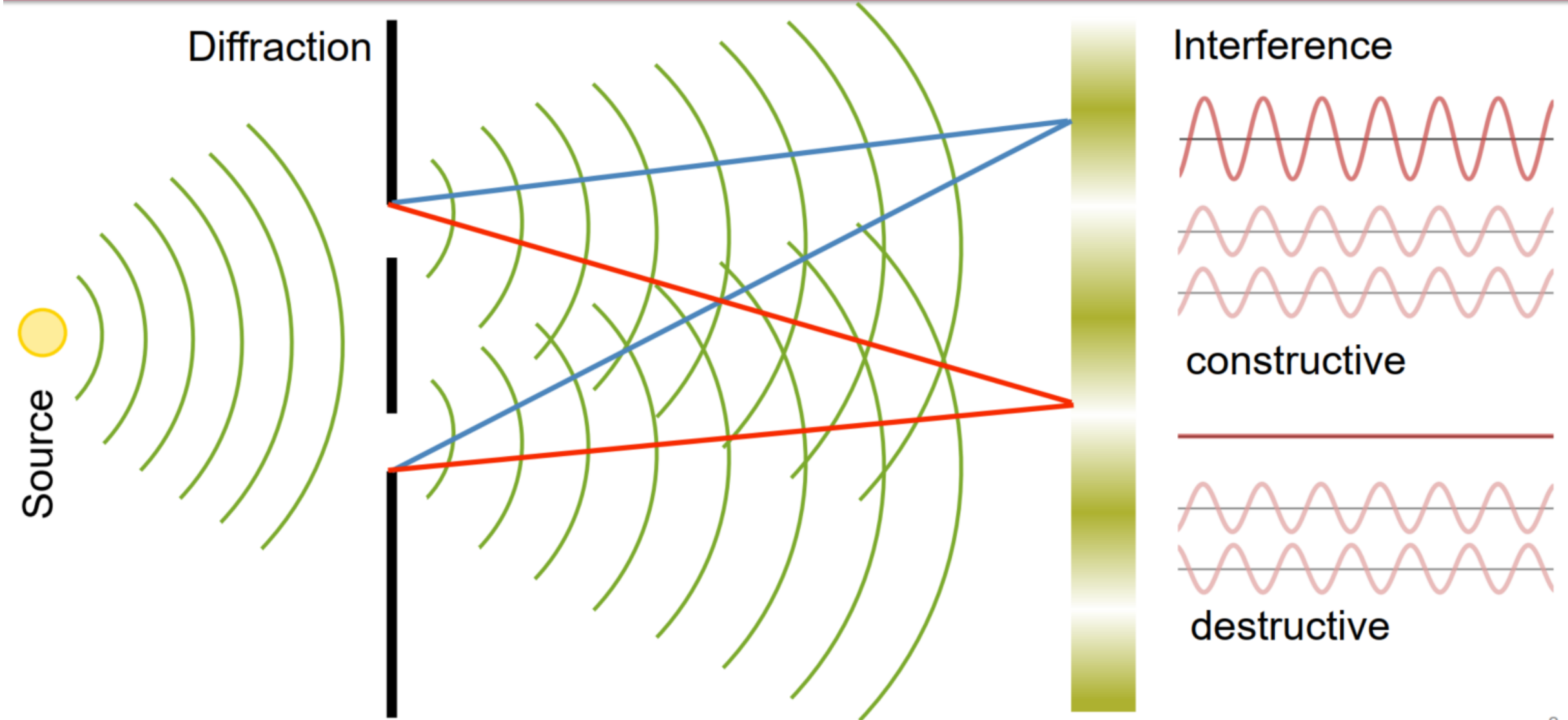
$$\vec{Q} = \vec{k}_i - \vec{k}_s$$

$$E_{tr} = \hbar \cdot \omega = \frac{\hbar^2}{2m} (k_i^2 - k_s^2)$$

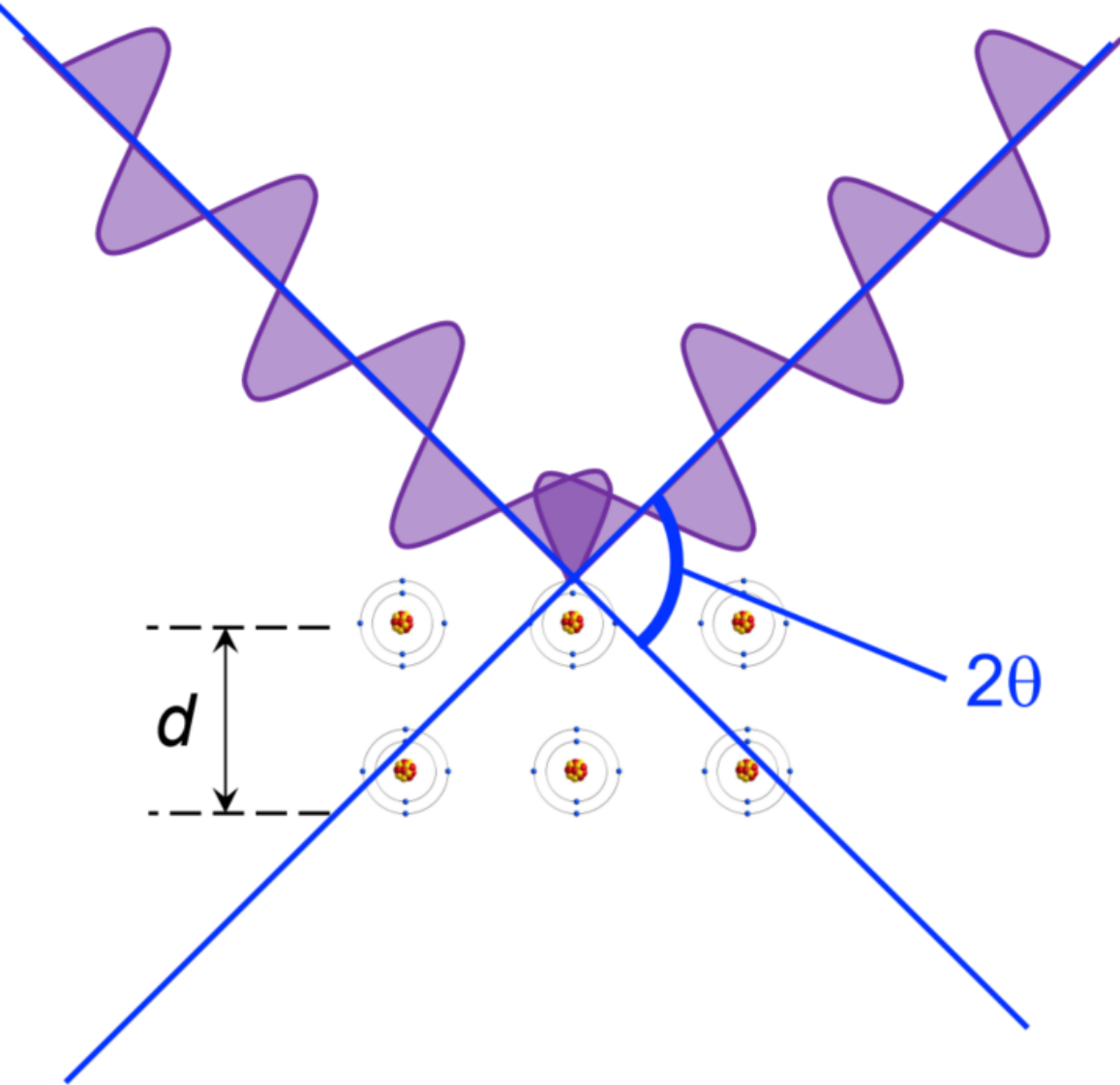
$E = 0$ ($|k_i| = |k_s|$): elastic scattering (diffraction)

$E \neq 0$ ($|k_i| \neq |k_s|$): inelastic scattering

Diffraction and Interference

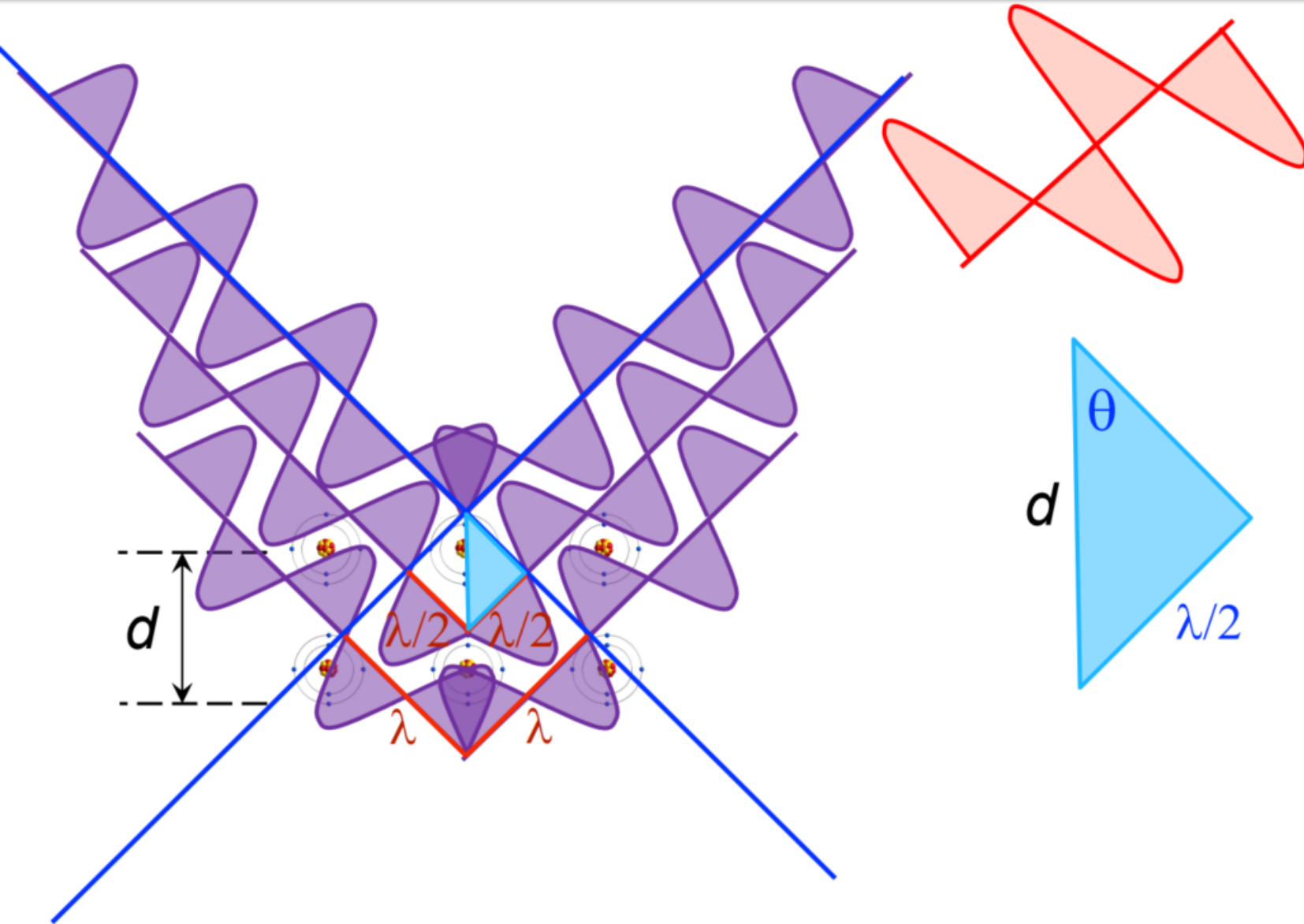


Bragg's law



In crystalline solids the repeating arrangement of atoms form distinct planes separated by well-defined distances acts like a **grating**

Bragg's law

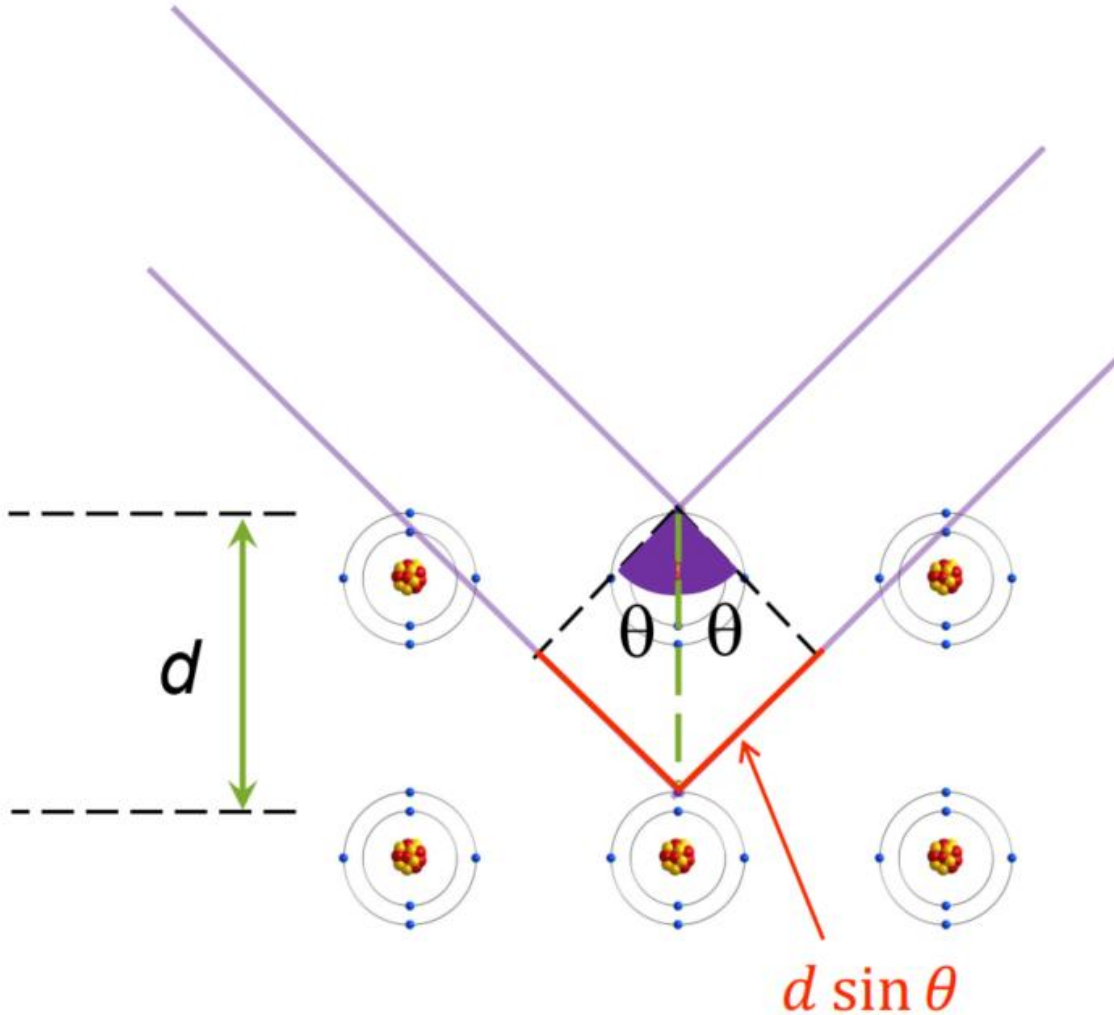


$$\sin \theta = \frac{\lambda}{2d}$$

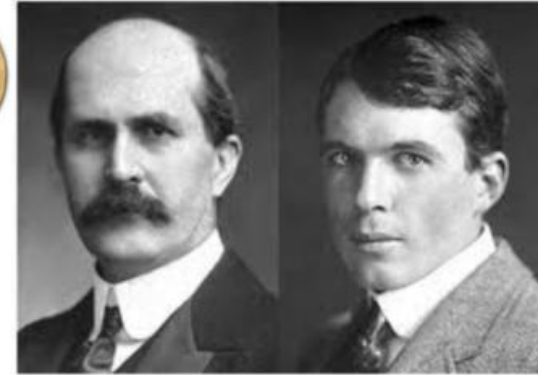
$$2d \sin \theta = n\lambda$$

Bragg's law

Bragg's law



1915



Henry
Bragg

Lawrence
Bragg

$$2d \sin \theta = n\lambda$$

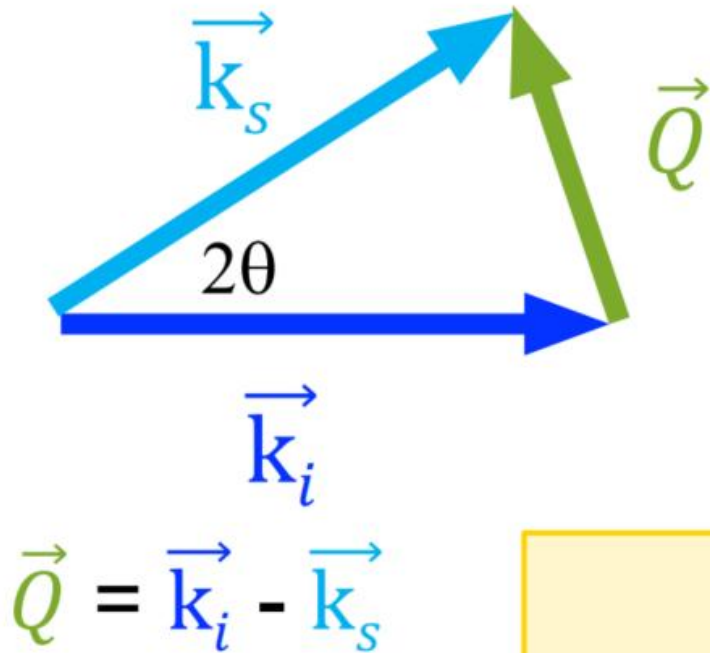
λ : wavelength

d : spacing between Miller plane

θ : angle between the incident beam and surface of crystal

Scattering triangle and Bragg law

Scattering triangle:



Elastic scattering

$$|k_i| = |k_s| = \frac{2\pi}{\lambda}$$

$$Q = 2k \sin \theta = 2 \sin \theta \cdot \frac{2\pi}{\lambda} = \frac{4\pi}{\lambda} \sin \theta$$

$$2d \sin \theta = n\lambda$$

$$Q = \frac{2\pi}{d} \quad \sin \theta = \frac{n\lambda}{2d}$$

We measure diffraction intensity as a function of angle or Q space.

☞ The diffraction is recorded in **reciprocal space**.

Different type of waves

X-ray diffraction (XRD)

- Daily used to provide structural information on compounds, identify samples and check the quality of sample
- Both powder and single crystal, Both in-house and synchrotron

Electron diffraction

- Phase identification, lattice parameter and space group determination
- Very small crystallites in the electron microscopy (TEM)

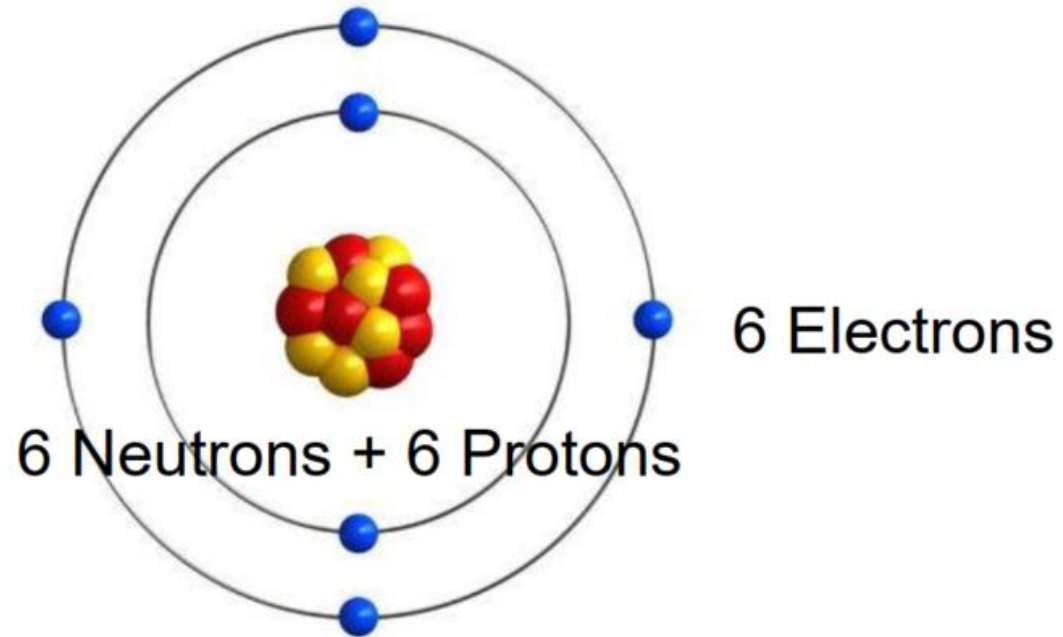
Neutron diffraction

- Complementary to X-ray for crystal structure determination
- Also possible to see magnetic ordering
- Non-destructive
- Need to be performed at large scale facilities
- Requires large volume of samples (powder and single crystal)

In-house neutron sources are under development but this dream will not come true until the end of your studies.

Different eyes: X-ray vs Neutron

How does a Carbon atom C^6 look like when probed with different types of radiation?

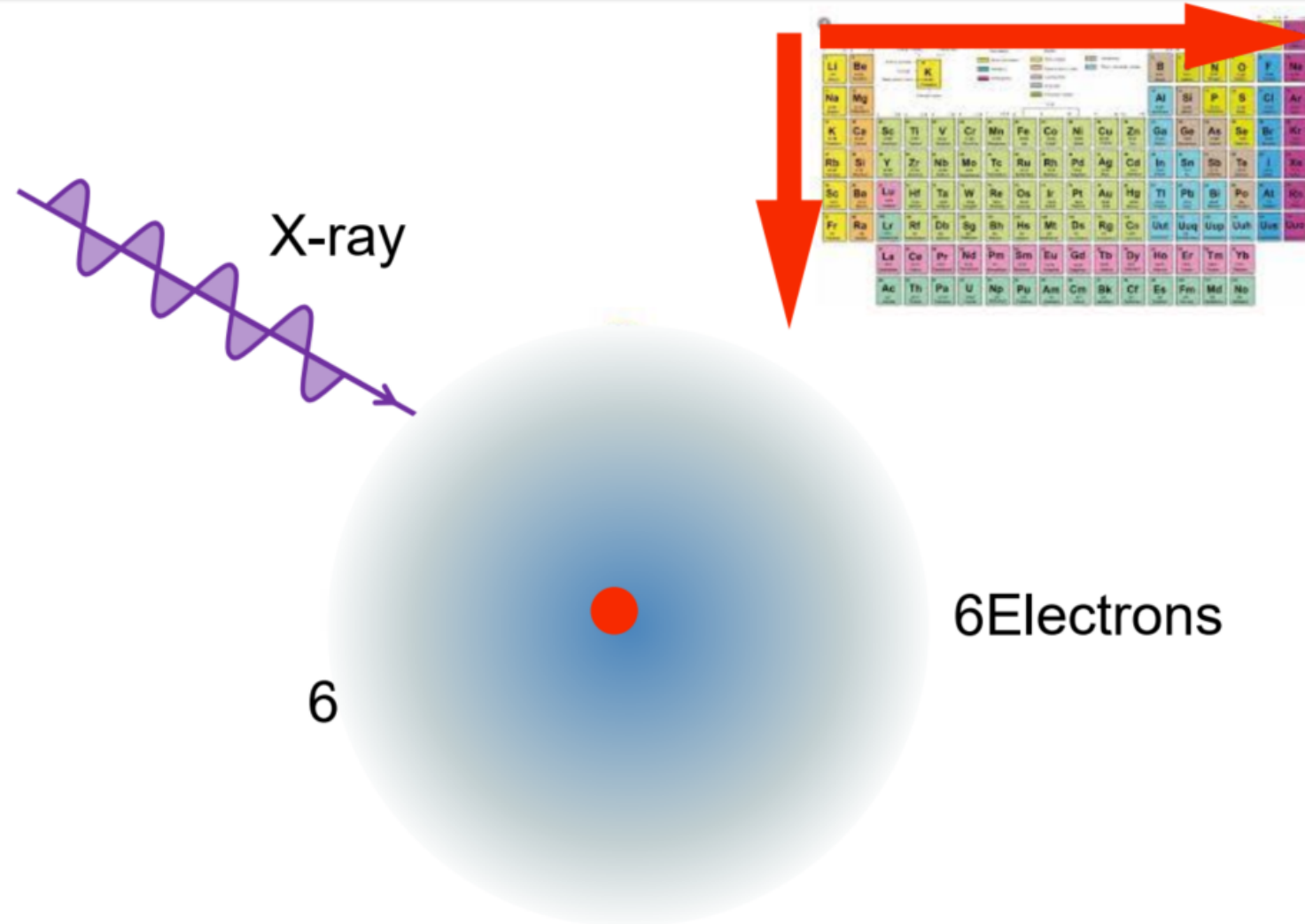


Different eyes: X-ray vs Neutron

Carbon atom C^6

X-ray

- ✎ X-ray has interactions with the electrons.
- ✎ X-rays are scattered by electron cloud of atoms.
- ✎ Form factor of X-ray has atomic number dependance.
- ✎ Small penetration depth
- ✎ Surface studies

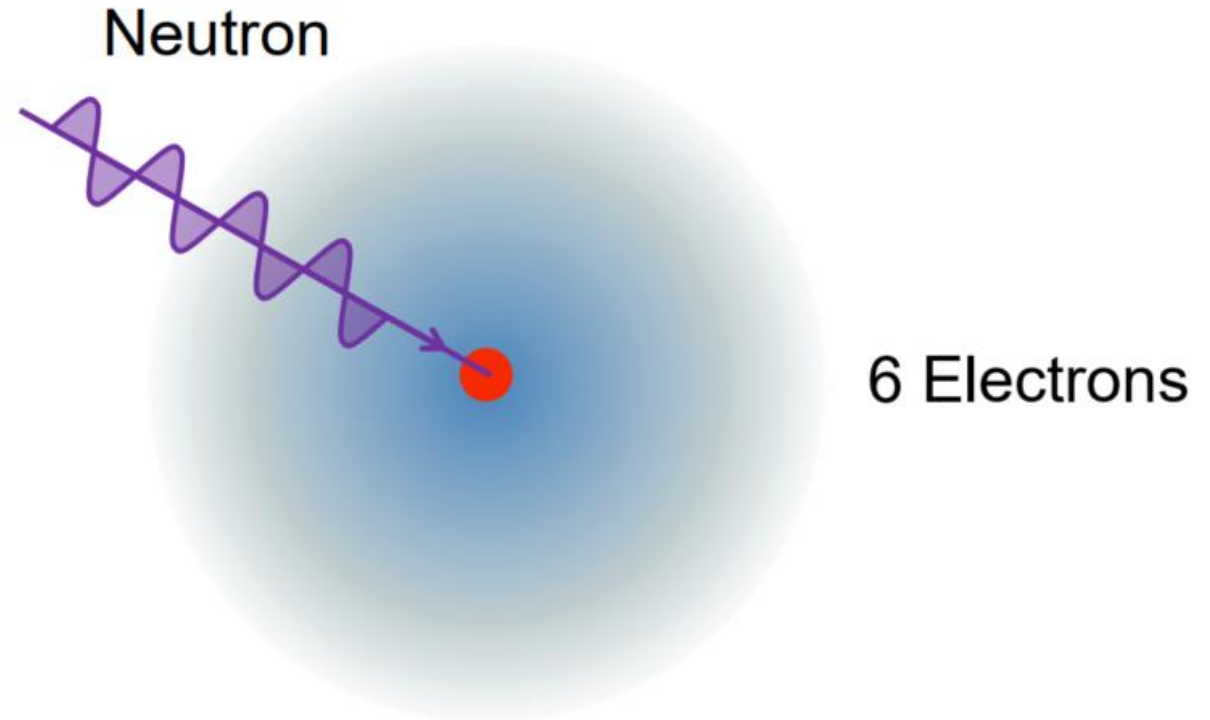


Different eyes: X-ray vs Neutron

Carbon atom C^6

Neutron (Structure)

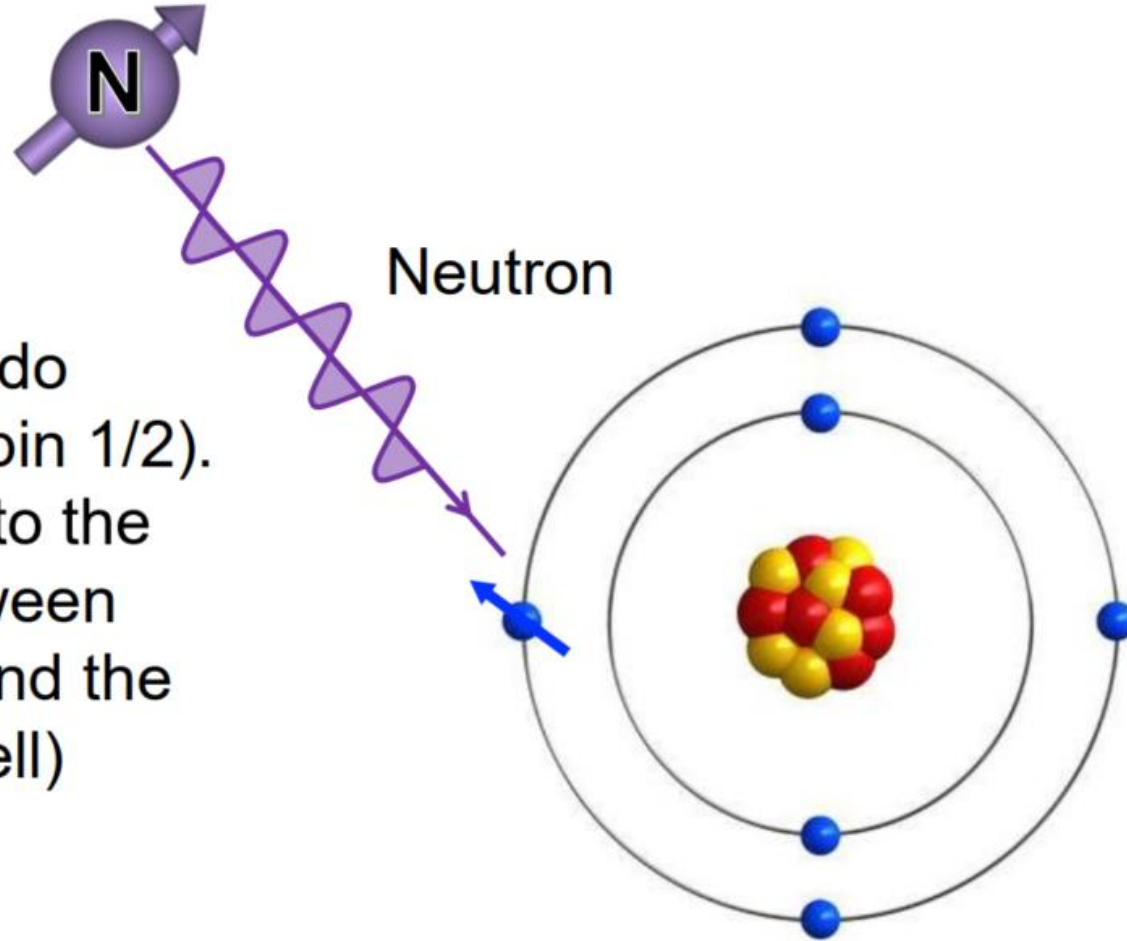
- Neutrons are uncharged and interact with the nuclei.
- Neutrons are scattered by nuclei (point-like target).
- Scattering length of neutron is independent on atomic number.
- Deep penetration capabilities
- Large volume of sample
- Bulk studies are possible



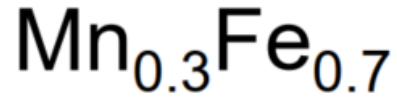
Different eyes: X-ray vs Neutron

Neutron (Spin)

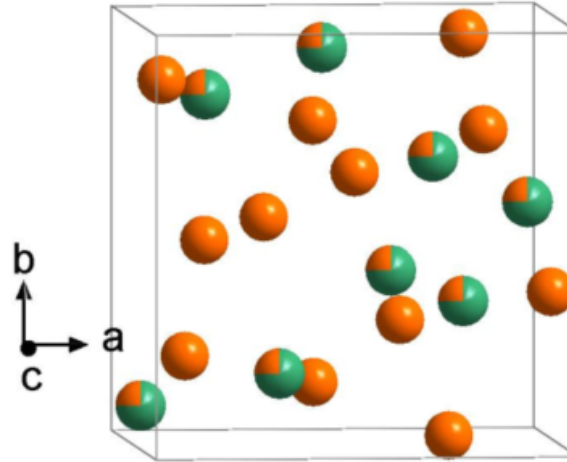
- Neutrons are uncharged but do have a magnetic moment (Spin $1/2$).
- An additional scattering due to the dipole-dipole interaction between the nuclear spin of neutron and the unpaired electrons (outer shell) occurs.



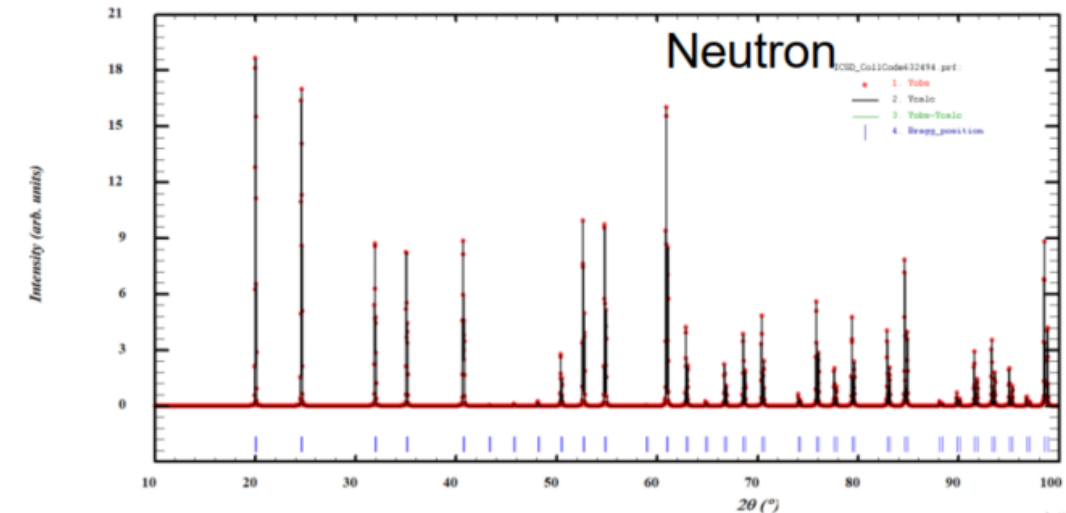
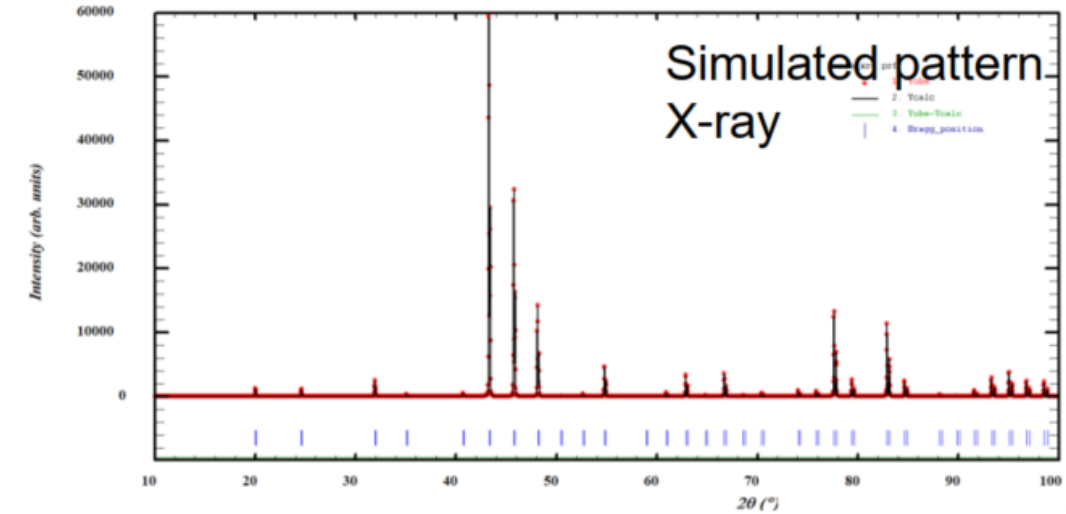
How different?: Example: $\text{Mn}_{0.3}\text{Fe}_{0.7}$



	Fe	Mn
Neutron	9.45	-3.73
X-ray (Z- number)	26	25



- ➡ More information at high-Q (high angles)
- ➡ The profile is simpler to analyze due to increased information
- ➡ The different scattering contrast among the elements gives large relative intensities

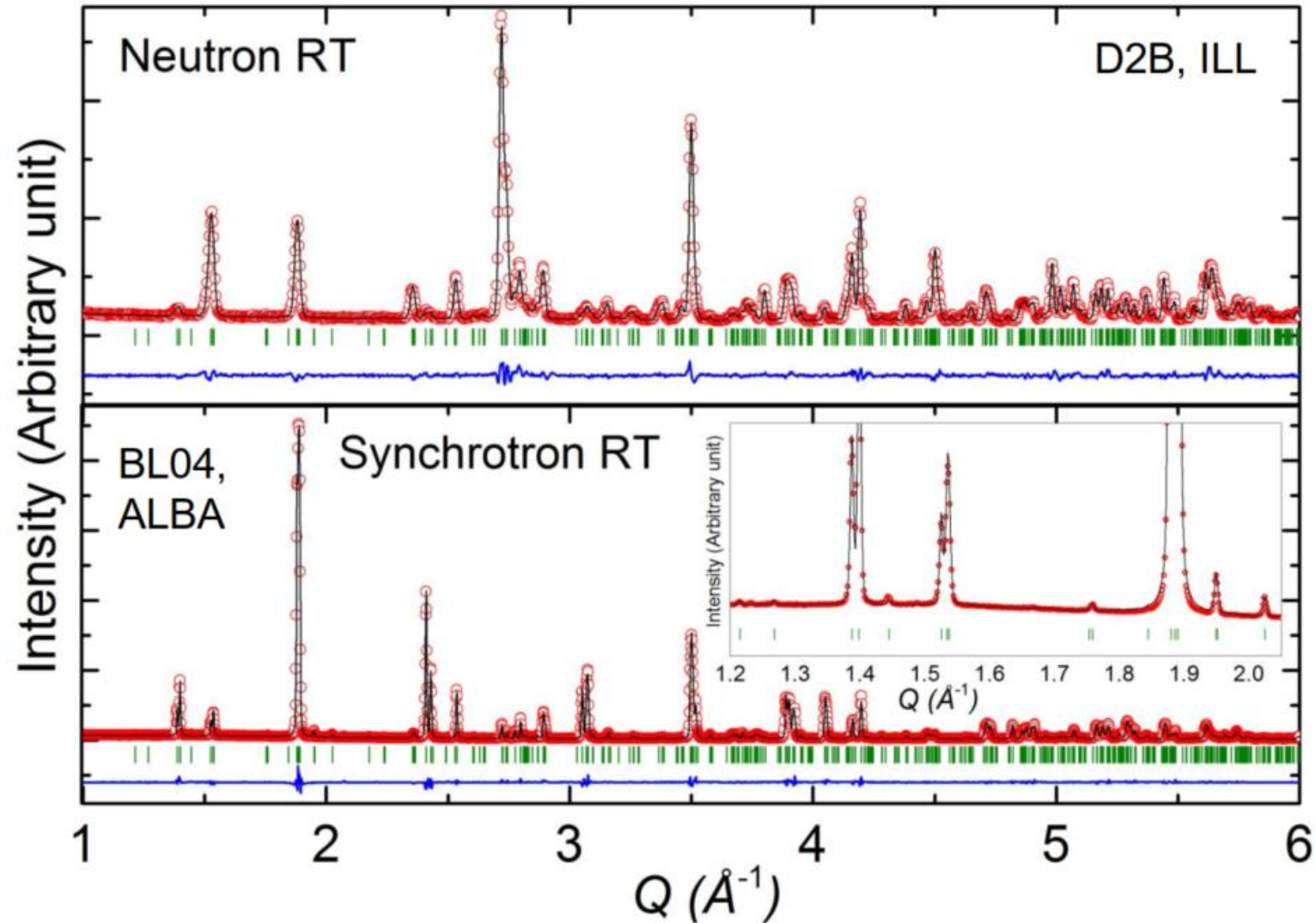


How different?: Example: BiMnTeO_6



	Bi	Mn	Te	O
Neutron	8.532	-3.73	5.8	5.8
X-ray (Z- number)	83	25	52	6

- ➡ More information at high-Q
- ➡ The profile is simpler
- ➡ Scattering contrast among the elements



Neutron vs X-ray: Practical

X-ray

- Daily used to provide structural information on compounds, identify samples and check the quality of sample
- Both powder and single crystal, Both in-house and synchrotron



Neutron diffraction

- In combo with X-ray, it is good complementary technique to determine crystal structure
- Allows to see magnetic ordering
- Non-destructive
- Need to be performed at large scale facilities
- Required large volume of samples (powder and single crystal)





Obrigado pela
sua atenção

